INSTITUTO COLLEGE

MATH 110: General Education Math

Fall 2026

Exam 1 Review

Overview of Topics

Counting

Sets

- Set, element: $x \in A$, $x \notin A$
- Cardinality: $|A|$; n-element set
- The empty set: $\emptyset$, with $|\emptyset| = 0$
- Subset: $A \subseteq B$
  - $A \subseteq A$ and $\emptyset \subseteq A$ for every set A
- Universal set U; complement A^c
- Union $A \cup B$; intersection $A \cap B$
- Disjoint sets: $A \cap B = \emptyset$
- $A \cap A^c = \emptyset$ and $A \cup A^c = U$
- Venn diagrams

The Fundamental Principle of Counting

- Tree diagrams
- The Fundamental Principle of Counting: $m_1 \cdot m_2 \cdot m_3 \cdot \ldots \cdot m_k$
  - Tossing a coin n times: 2^n possible sequences
  - Rolling two dice: $6 \cdot 6 = 36$ possible results

Factorials

- $0! = 1$ and $n! = n \cdot (n-1)!$
- $n! = n(n-1)(n-2) \cdot \ldots \cdot 3 \cdot 2 \cdot 1$

Permutations

- Permutation: an arrangement in which order matters
- Number of permutations of n elements: $n!$
- k-permutation; permutation number: ${}_nP_k = \dfrac{n!}{(n-k)!}$

Combinations

- Combination: a choice in which order does not matter
- k-combination; combination number: ${}_nC_k = \dfrac{n!}{k!\,(n-k)!}$
- ${}_nP_k = {}_nC_k \cdot k!$

Counting Arguments

- Using the FPC, permutations, and combinations together
- The Addition Principle: If A and B are disjoint, then $|A \cup B| = |A| + |B|$. More generally, if A_1, A_2, …, A_k are disjoint, then

$$|A_1 \cup A_2 \cup \ldots \cup A_k| = |A_1| + |A_2| + \ldots + |A_k|.$$

- Counting with complements: $|A| = |U| - |A^c|$
- The Inclusion-Exclusion Principle:

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

Probability

Sample Spaces and Events

- Experiment; outcome, or sample point; sample space S
  - A coin tossed n times: S is the set of length-n sequences of H's and T's. $|S| = 2^n$.
  - Two dice: S is the set of ordered pairs of integers (m, n) with $1 \le m, n \le 6$. $|S| = 36$.
  - The blood type experiment: $S = \{\text{A}+, \text{A}-, \text{B}+, \text{B}-, \text{AB}+, \text{AB}-, \text{O}+, \text{O}-\}$
- Event: a subset of S

Probability by Counting

- Sigma notation: $\sum_{x \in S} P(x)$ means "the sum of the probabilities of all the sample points in S."
- Each sample point x in S is assigned a probability $P(x)$, and these numbers must satisfy two conditions:
  1. $0 \le P(x) \le 1$ for every sample point x in S
  2. $\sum_{x \in S} P(x) = 1$
- Probability of an event
  - $P(A) = \sum_{x \in A} P(x)$
  - $P(S) = \sum_{x \in S} P(x) = 1$
  - $P(\emptyset) = 0$

- Equally likely sample points
  - $P(x) = \dfrac{1}{|S|}$
  - Probability by Counting: $P(A) = \dfrac{|A|}{|S|}$

Mutually Exclusive Events and the Addition Rule

- Mutually exclusive events: $A \cap B = \emptyset$
- The Addition Rule: If A and B are mutually exclusive, then $P(A \cup B) = P(A) + P(B)$. More generally, if A_1, A_2, ..., A_k are mutually exclusive, then

$$P(A_1 \cup A_2 \cup \ldots \cup A_k) = P(A_1) + P(A_2) + \ldots + P(A_k).$$

- Probabilities of complements: $P(A^c) = 1 - P(A)$
- The Inclusion-Exclusion Principle for Events: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Independence and the Multiplication Rule

- Independent events: $P(A \cap B) = P(A) \cdot P(B)$
- The Multiplication Rule: when independence is given, multiply
  - Tossing a coin several times, rolling several dice, drawing with replacement
  - Extends to three or more independent events
- Drawing without replacement: the draws are not independent

Conditional Probability

- Conditional probability: $P(A \mid B) = \dfrac{P(A \cap B)}{P(B)}$
- Equally likely sample points: $P(A \mid B) = \dfrac{|A \cap B|}{|B|}$
- Independence: $P(A \mid B) = P(A)$ and $P(B \mid A) = P(B)$
- Mutual exclusion vs. independence – not the same!
- Two-way tables
- $P(A \mid B)$ and $P(B \mid A)$ are not the same
- The General Multiplication Rule: $P(A \cap B) = P(B) \cdot P(A \mid B) = P(A) \cdot P(B \mid A)$

Binomial Experiments

- Binary experiment, or Bernoulli trial: success or failure
- Binomial experiment
  - The number of trials n is fixed
  - The trials are independent
  - The probability of success p is the same on every trial
- Binomial probabilities: $P(E_k) = {}_nC_k \cdot p^k \cdot (1-p)^{n-k}$, where E_k is the event "exactly k successes"